Product Requirements

AstroIllustrator | Consolidated specification v2.1 | 22 October 2026

1. Product vision

Create scientifically accurate, publication-ready astronomy illustrations for a commercial Albanian/English atlas. One application supports sky charts, mathematical/scientific diagrams and optional artistic composites. AI helps interpret requests and generate scenery; it is never the authority for celestial coordinates or measured angles. Primary user: atlas author/editor. Secondary users may be educators. Local-first single-user workflow; no accounts, billing or SaaS requirements in v2.

2. Illustration families

SkyChart: SingleConstellation, MultipleConstellations, Asterism, HorizonView, SeasonalSky, AllSky. Asterisms are distinct from official IAU constellations (Summer Triangle: Vega, Deneb, Altair). ScientificDiagram: StarAltitude, CelestialSphere, AltitudeAzimuth, EquatorialCoordinates, AngularSeparation, PolarisLatitude, EarthSeasons. Later templates may cover Moon phases, parallax, eclipses and Kepler geometry. ArtisticIllustration: standalone decorative background or ScientificComposite; the accurate chart/diagram remains a separate locked vector layer.

3. Representative use cases

UC-01: Generate Orion from verified catalog and constellation line definitions, export SVG/PNG. UC-02: Generate Lyra, Cygnus and Aquila with a Summer Triangle asterism overlay. UC-03: Render southern horizon from Prishtina for a chosen UTC date/time, with correct local altitude/azimuth and visibility filtering. UC-04: Generate an annotated Polaris altitude diagram from an observer latitude, optionally resolving exact altitude for a supplied time. UC-05: Enter a prompt such as “Show Polaris from Prishtina with its angle above the northern horizon, in Albanian”; model returns a validated diagram specification; scientific engine computes the result. UC-06: Create an atlas figure with chapter/figure ID, style preset, print dimensions and reproducible manifest.

4. Essential authoring capabilities

Provide structured/manual configuration even when AI is offline. Support observer latitude, longitude, elevation, UTC timestamp, projection, field of view, orientation, limiting magnitude, labels, grid and horizon visibility. Offer consistent presets: Dark Atlas, Clean Scientific, Educational Diagram, Print Monochrome and Scenic Composite. Support Albanian and English terminology with editable labels and caption fields. Exports: SVG first, vector PDF when feasible, high-resolution PNG for preview. Keep output metadata and illustration manifest.

5. MVP boundary and phases

MVP-A: accurate Orion and Ursa Minor charts, SVG/PNG, presets and validation. MVP-B: multi-constellation, Summer Triangle, observer/time and southern horizon. MVP-C: deterministic zenith/altitude/azimuth and Polaris diagrams. AI prompt-to-typed-request and AI landscape backgrounds are subsequent enhancements, not dependencies of the scientific MVP. RAG and tool-calling assistant are optional later stages.

6. Acceptance criteria

Astronomical points and angles derive from traceable calculations; no AI-created star geometry. Summer Triangle connects the correct identified stars. Horizon output respects observer, time, projection and visibility. Polaris demo distinguishes the north celestial pole altitude (equal to observer latitude on an ideal horizon) from the star Polaris, whose exact altitude changes slightly with time. Exports preserve measured geometry across visual presets; labels are readable at intended print dimensions; each image can be regenerated from a stored manifest. Document control: This version supersedes earlier 1.0 and 1.1 scope descriptions. Keep requests, manifests, diagrams and AI integrations aligned with this specification.

v2.1 additions: global observer support, time-aware conjunctions and AstroIllustrator naming

Global observing locations

All computed sky views and astronomical event illustrations support arbitrary Earth coordinates, not a Prishtina-only default. Users can select cities, manually enter WGS84 latitude/longitude and elevation, save presets, and select local civil time or UTC. Latitude is north/south (-90 to +90); longitude is east/west (-180 to +180). Prishtina (about 42.66 N, 21.16 E) is merely a sample preset.

Astronomical event illustrations

Add SkyChart scene type AstronomicalEvent with event categories PlanetaryConjunction, MoonPlanetConjunction, PlanetaryAlignment and TwilightVisibility. Users choose target bodies, date or date range, observing location, viewing direction (e.g., western horizon), and an absolute or sunset/sunrise-relative time. Include angular separation, horizon, twilight shading and visibility notices.

New author use cases

UC-E01: render Venus and Jupiter above the western horizon 30 minutes after sunset at a chosen city/date; verify whether the specified date actually contains a close pairing. UC-E02: show a crescent Moon and Venus after sunset with correct lunar phase. UC-E03: compare the same event from two locations. UC-E04: search a date interval for closest apparent approach and choose a render time.

Event acceptance criteria

No fabricated conjunctions: report calculated angular separation and distinguish close visual pairing from exact conjunction terminology. For a body below the horizon, mark not visible rather than moving it. Support no-sunset days at high latitude; honor UTC/time zone/DST; offer exact reproducible event metadata. Rendering from southern hemisphere must work.

Scientific note: illustrative dates and prompts are examples, not verified event predictions. All published charts require computed ephemerides and editorial review.